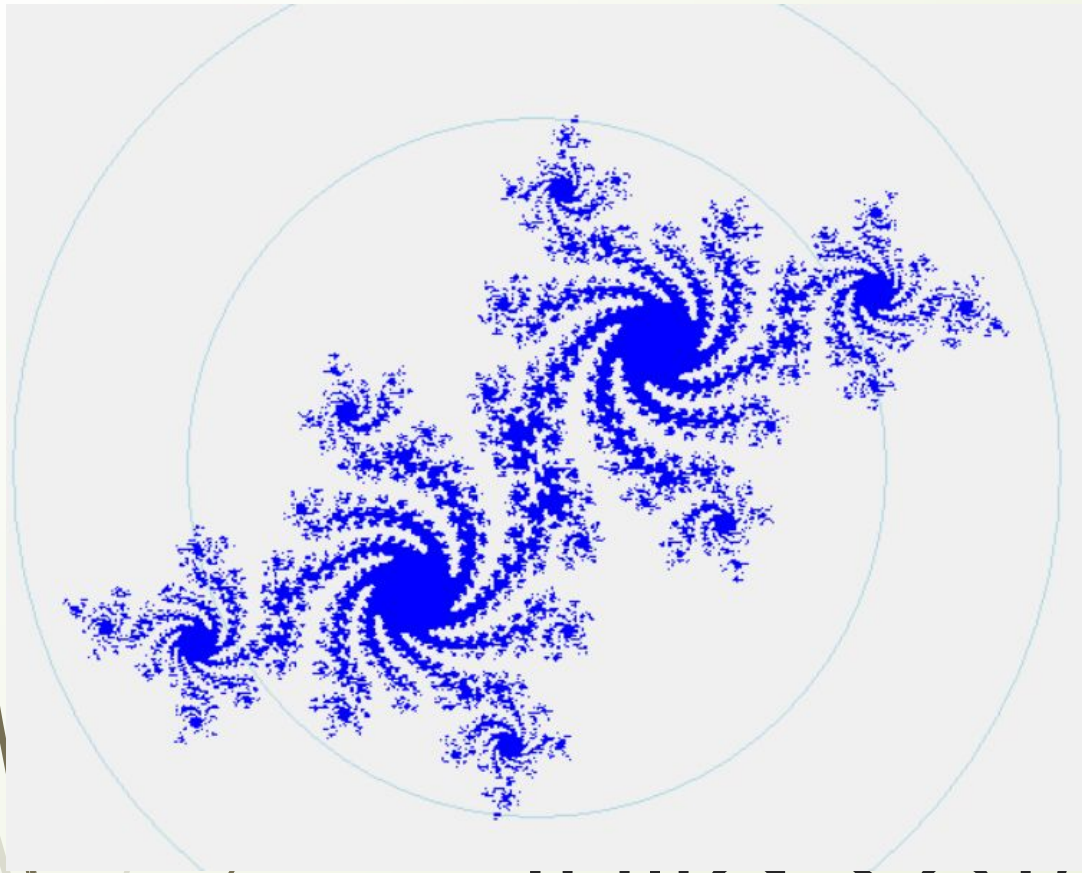
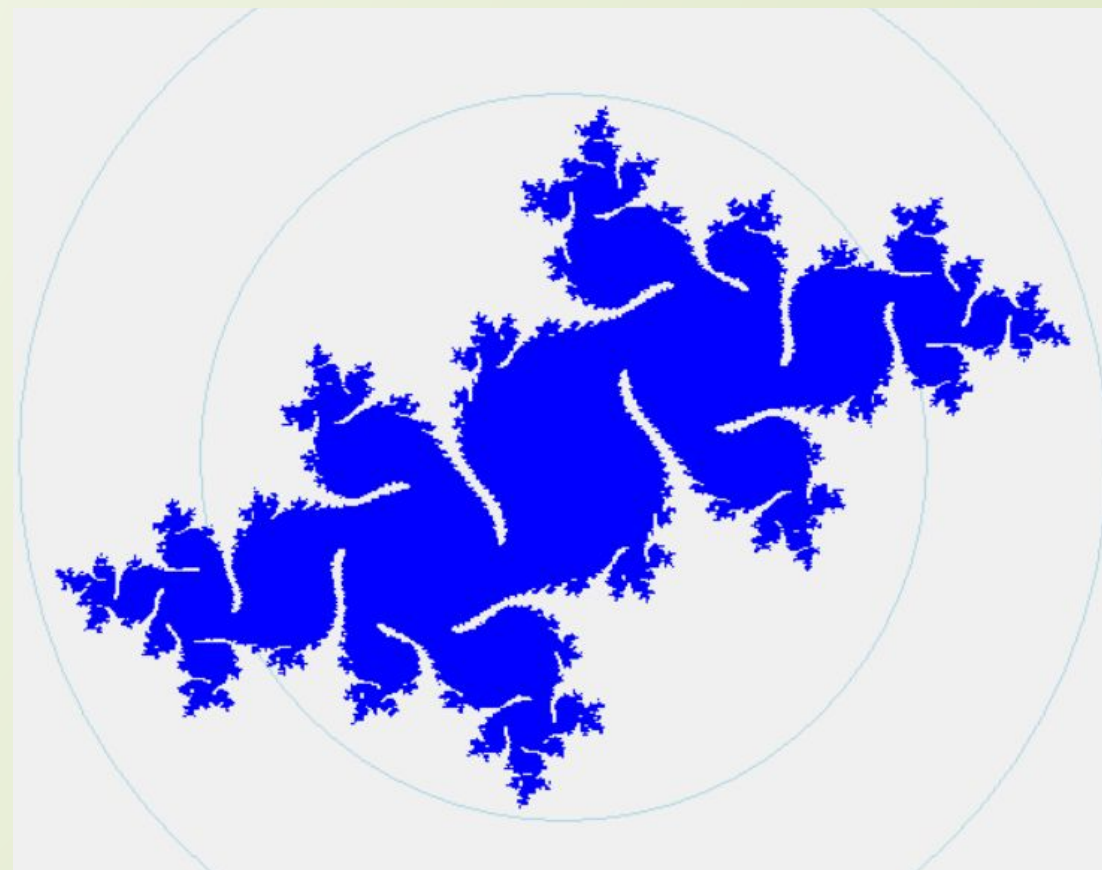




JULIA SETS

“Julia Lace” or filled sets

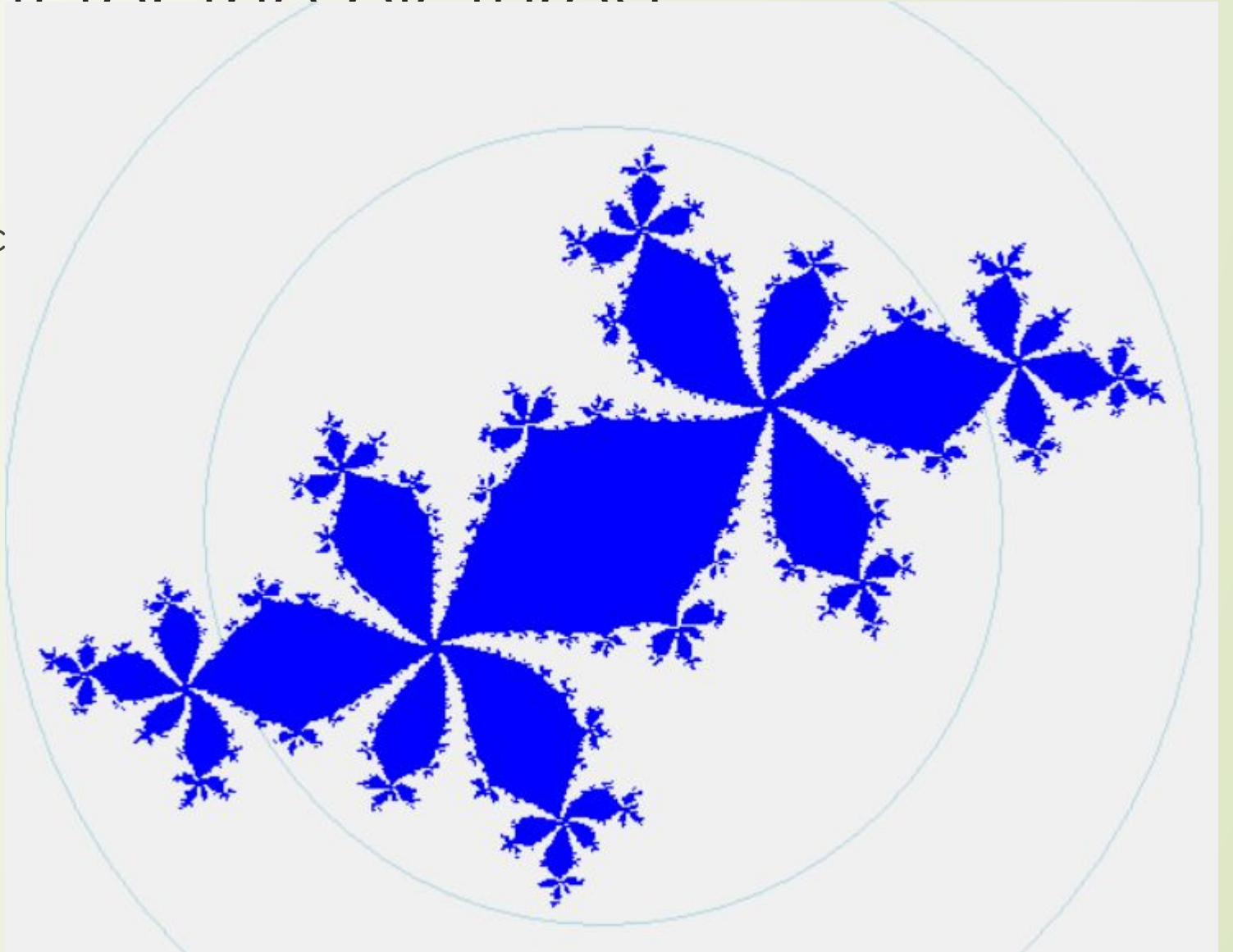


Background – via Wikipedia

- These sets are named after the French mathematicians Gaston Julia^[1] and Pierre Fatou^[2] whose work began the study of complex dynamics during the early 20th century.
- In complex dynamics, the **Julia set** and the Fatou set are two complementary sets (Julia "laces" and Fatou "dusts") defined from a function. Informally, the Fatou set of the function consists of values with the property that all nearby values behave similarly under repeated iteration of the function, and the Julia set consists of values such that an arbitrarily small perturbation can cause drastic changes in the sequence of iterated function values. Thus the behavior of the function on the Fatou set is "regular", while on the Julia set its behavior is "chaotic".

I'm just in it for the pictures!

□ Yes, they're frac





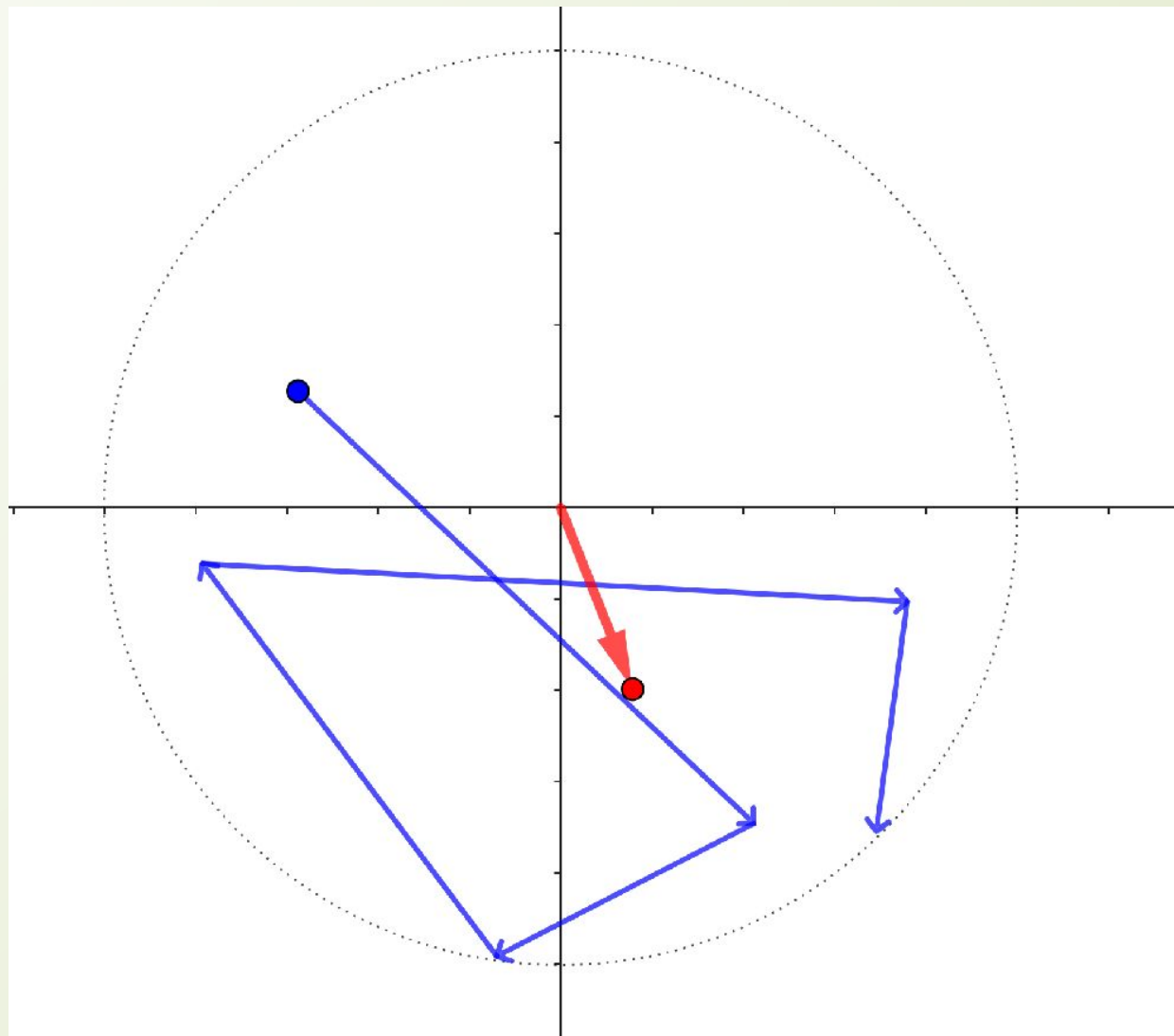
Bounded and Unbounded under iteration

- Given a complex-valued function, a complex point produces an infinite sequence under iteration.
- Points which produce unbounded sequences are called “unbounded under iteration”, or just “unbounded”.
- Points which are not unbounded are “bounded under iteration”.
- The “filled Julia set” is the set of all bounded points.
- The Julia set is the set of border points between the bounded and unbounded.
 - Hence, they are “chaotic” or “twitchy”.

Equivalent Descriptions – via Wikipedia

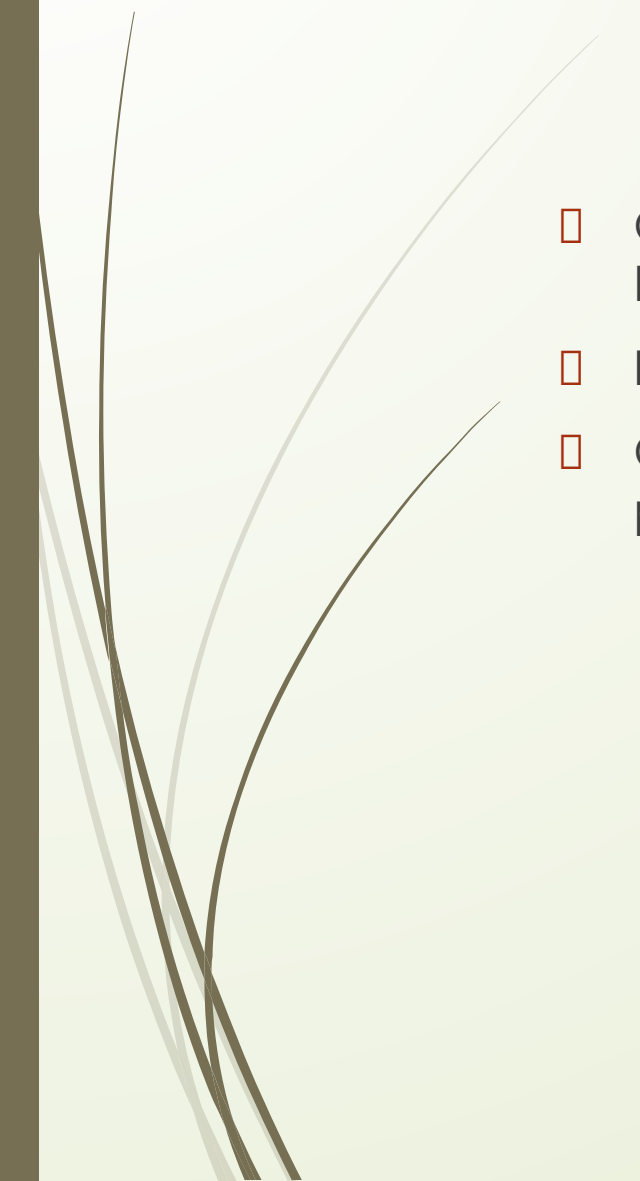
- ▣ $J(f)$ is the smallest closed set containing at least three points which is completely invariant under f .
- ▣ $J(f)$ is the closure of the set of repelling periodic points.
- ▣ For all but at most two points $z \in X$, the Julia set is the set of limit points of the full backwards orbit $\bigcup_n f^{-n}(z)$.
- ▣ If f is an entire function, then $J(f)$ is the boundary of the set of points which converge to infinity under iteration.
- ▣ **If f is a polynomial, then $J(f)$ is the boundary of the filled Julia set; that is, those points whose orbits under iteration remain bounded.**

Huh? Time for a Geogebra demo...

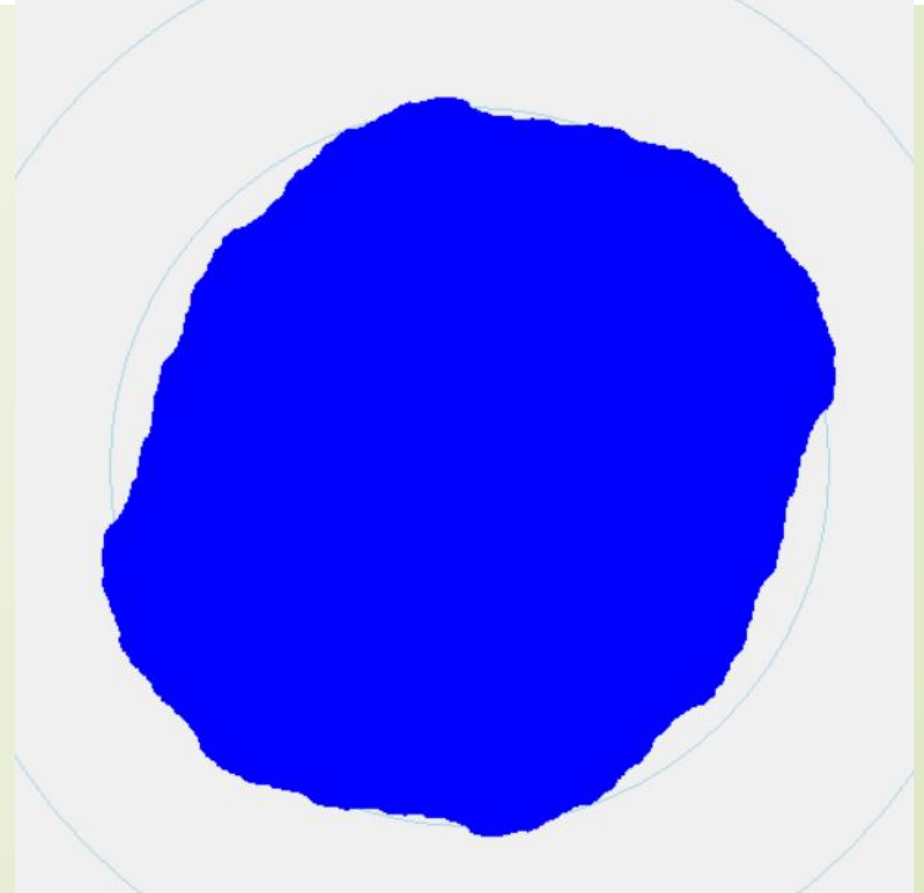
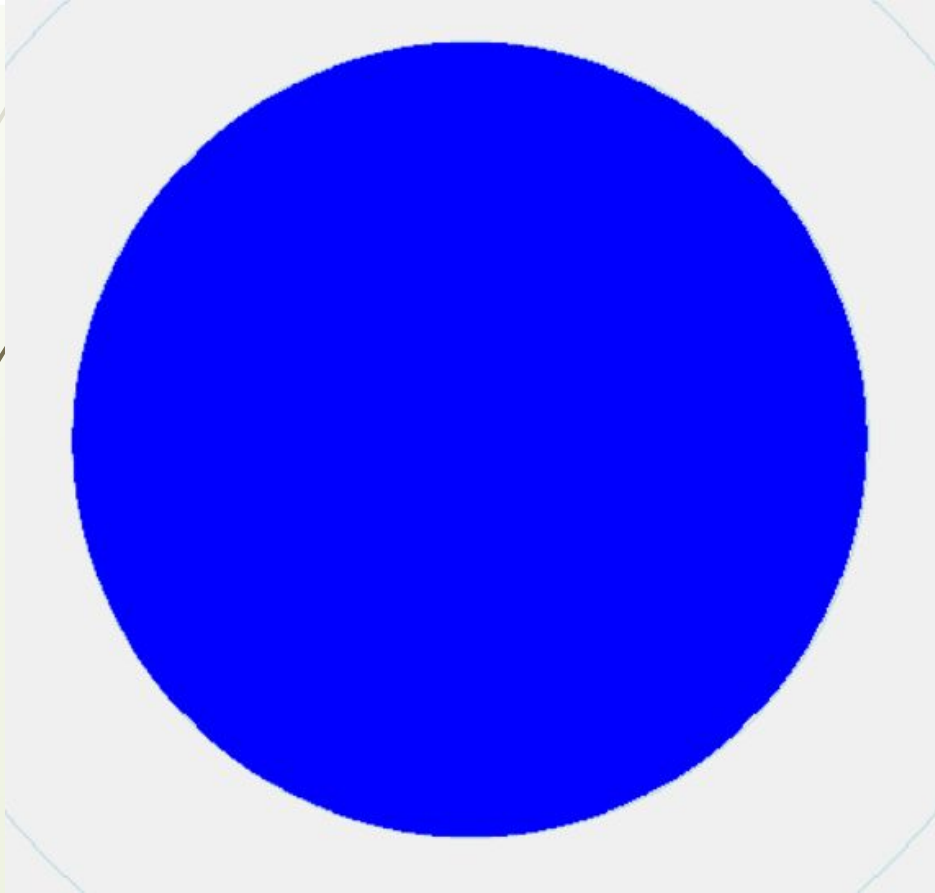




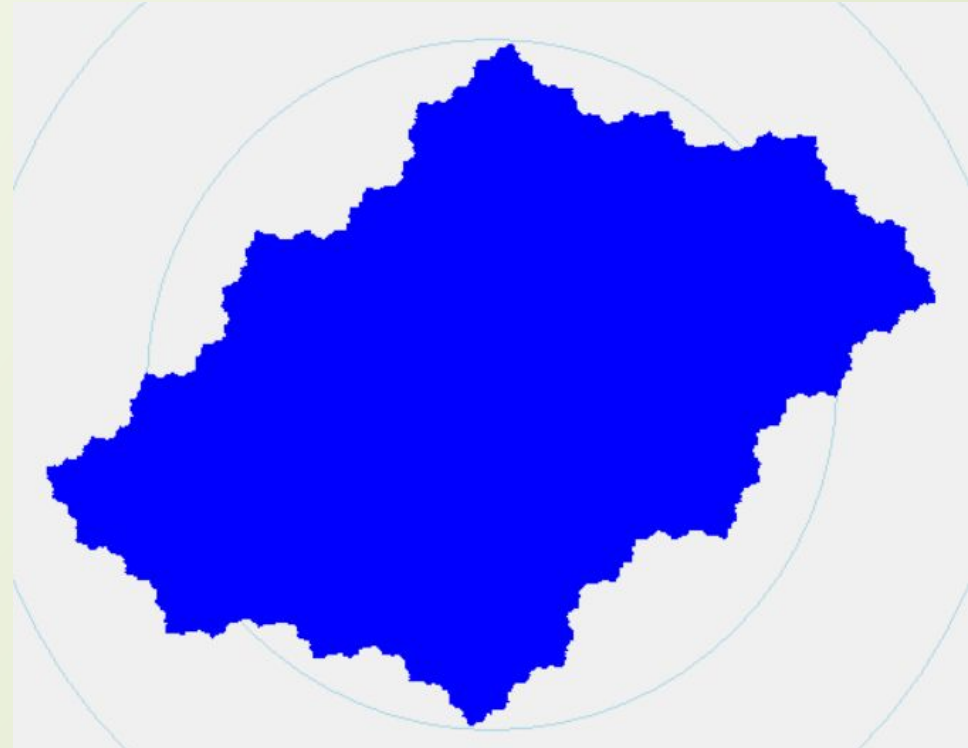
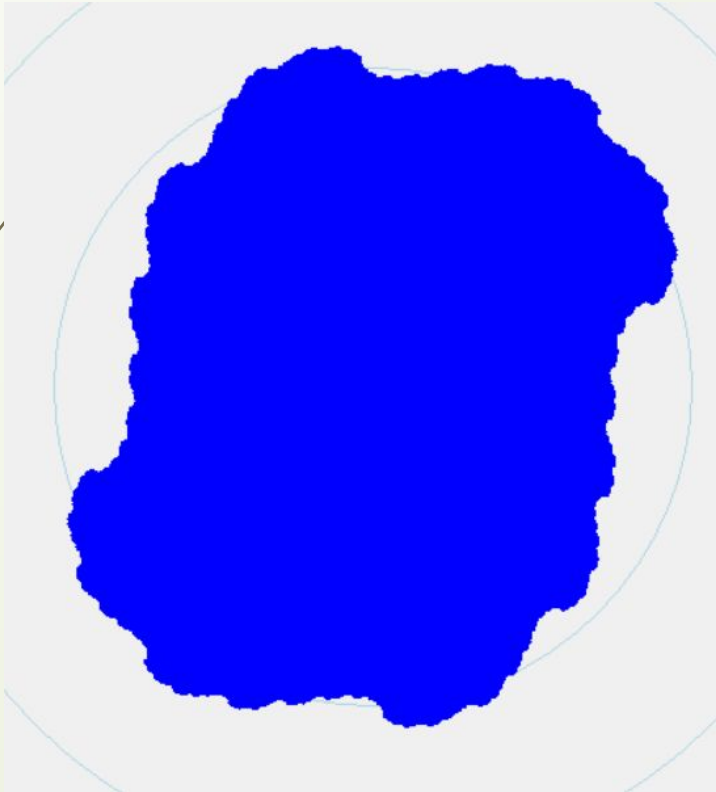
Observations – Initial Point & Constant

- Constants with radius greater than about 1.2 tend to have very few bounded points, if any.
 - Initial points with radius greater than 1.6 are likely to be unbounded.
 - Constants with radius between 0.6 and 1.0 produce some of the best pictures!
- 

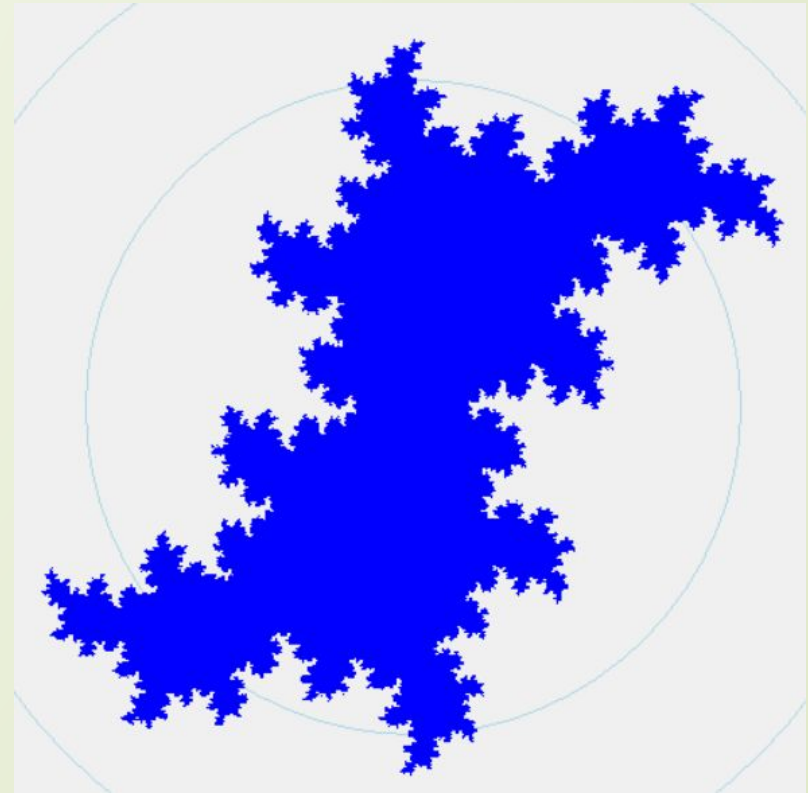
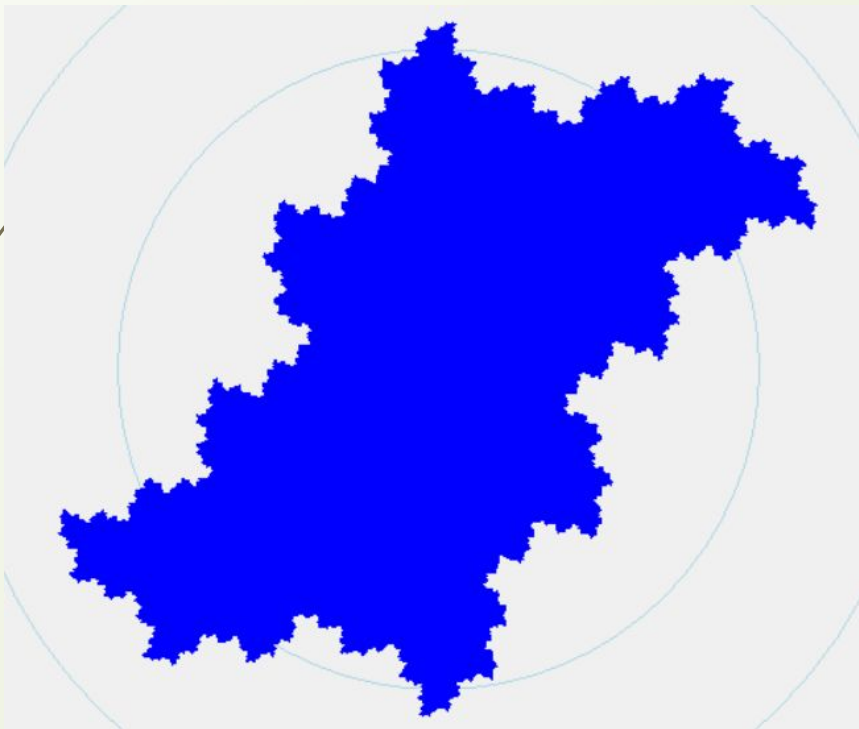
Boring!



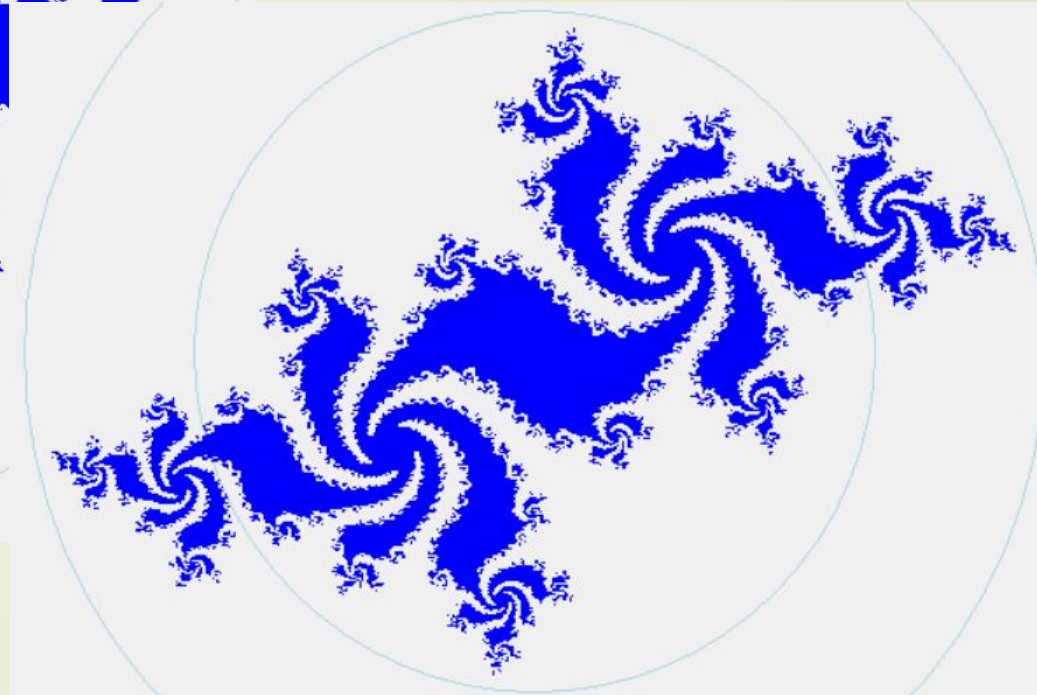
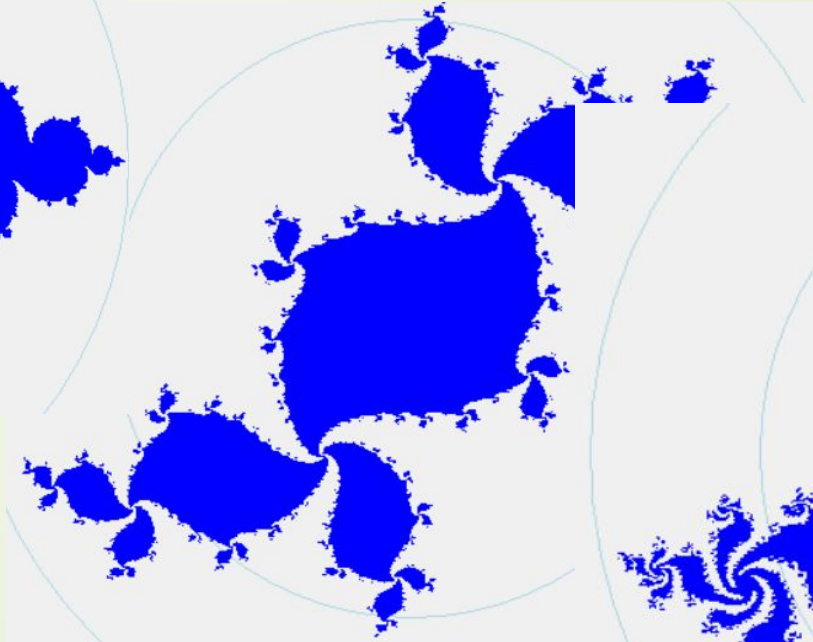
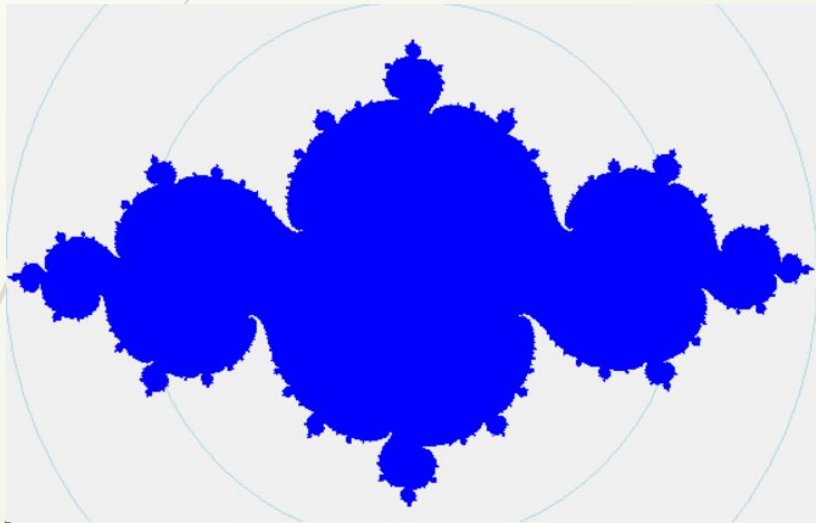
Meh...



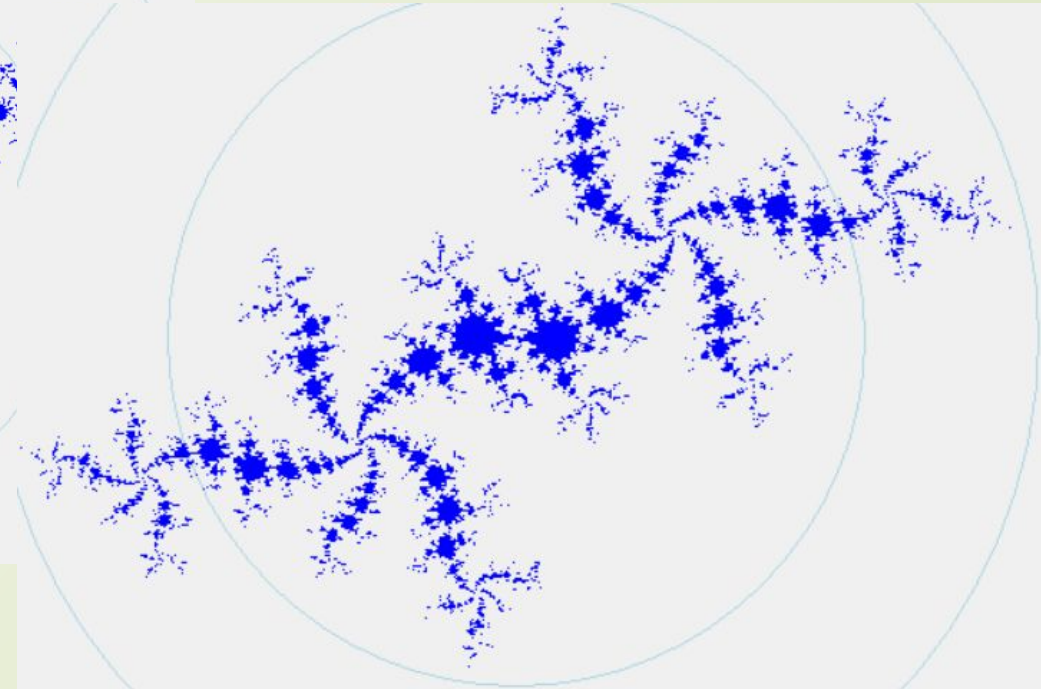
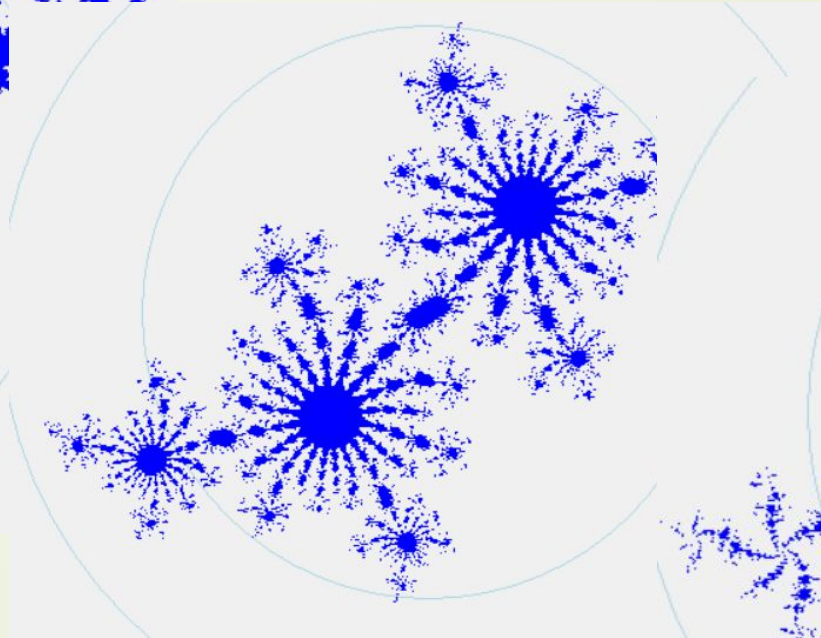
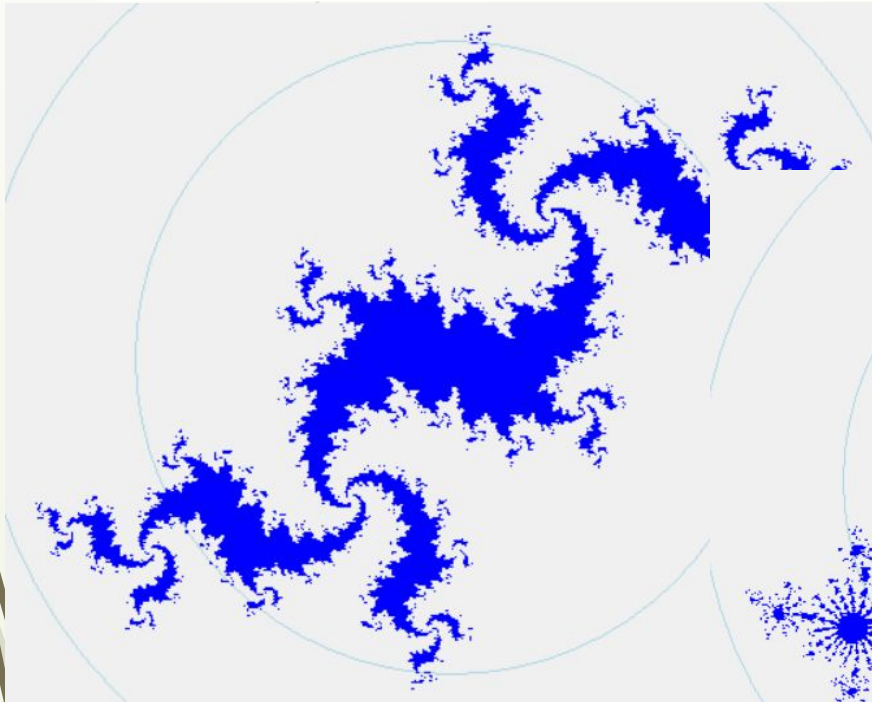
Cooler...



➤ Weirder...



Disappearing...



Fun facts!?

- ➡ If z is a point in a Julia set, so is $-z$. So the Julia set for $z^2 + C$ has 180° symmetry.
 - ➡ What symmetry would $z^3 + C$ or $z^4 + C$ have?
- ➡ A function $f(z) = z^2 + C$ has two fixed points at $z = \frac{1 \pm \sqrt{1-4C}}{2}$.
- ➡ When $C = -1$,
 - ➡ We get a repeating sequence $-1, 0, -1, 0, \dots$
 - ➡ One of the boundary (Julia) points is $\frac{1+\sqrt{5}}{2}$, the Golden Mean.

Fireworks, anyone?





A link to more pretty...

- <https://icefractal.com/julia/>
 - https://en.wikipedia.org/wiki/Julia_set
- 